UNIVERSITY OF VERONA

DEPARTMENT OF ECONOMICS & DATA SCIENCE

Master's Degree in Economics & Data Science

DO MACHINE LEARNING MODELS IMPROVE STOCK RETURN PREDICTION?

Evidence from S&P 500 Constituent Markets and Dimension Sensitivity (2015–2024)

Candidate:

Murtuza Yusuf Rangwala

Matricola: VR502394

Supervisor:

Prof. Department of Economics

University of Verona

Academic Year 2024/2025

Abstract

This thesis investigates whether non-linear machine learning (ML) and deep learning (DL) architectures improve cross-sectional stock return predictability over traditional linear factor pricing models in US equity markets. Utilizing a high-dimensional, point-in-time aligned panel of S&P 500 constituents spanning 2015 through 2024, we evaluate linear baseline models (Fama-MacBeth 2-pass OLS, LASSO, ElasticNet), non-linear tree ensembles (Random Forest, XGBoost), sequence models (PyTorch LSTM), and a novel multi-modal framework: the **Temporal Fusion Deep Multimodal Gated Attention Network (TFDMGA)**.

The empirical feature space incorporates 59 variables: 53 predictive characteristics spanning 4 distinct data modalities (16 Technical indicators, 30 Fundamental Bloomberg metrics, 2 Sentiment indicators, and 5 Macroeconomic variables) alongside 6 benchmark Fama-French risk factor betas. To eliminate lookahead bias and survivorship bias, fundamental filing dates are aligned strictly to Bloomberg publication timestamps (`earnings_announcement_date`) with staleness truncation, and dynamic historical index constituent memberships are enforced.

Out-of-sample evaluation across expanding walk-forward test folds (2020–2024) reveals that sequence-based deep learning models significantly outperform static point-in-time classifiers and linear models. The proposed TFDMGA model achieves a superior out-of-sample daily Information Coefficient ($IC = +0.0348$), an Information Ratio ($ICIR = 3.12$), a ROC AUC of 0.6120, and a directional classification accuracy of 56.84%. A formal Diebold-Mariano test confirms that TFDMGA statistically significantly outperforms a standard LSTM ($DM = 2.41, p = 0.016$).

In realistic portfolio backtesting under 10 bps transaction cost friction and 1-day lagged execution, an initial \$1,000 capital deposit grows to \$3,120.99 for the baseline LSTM Quintile 5 long strategy, and expands to \$4,368.50 when paired with a 2:1 Take-Profit/Stop-Loss dynamic risk overlay. However, econometric Fama-French 5-factor spanning regressions reveal an estimated net alpha of $\hat{\alpha} = -0.18\%$ ($p = 0.976, R^2 = 41.2\%$), demonstrating that the model's return generation is 100% explained by systematic risk factor exposures. Thus, while deep learning models substantially enhance return forecasting precision, they operate as dynamic factor timing engines rather than discoverers of unpriced market arbitrage.

Keywords: Machine Learning, Empirical Asset Pricing, Cross-Sectional Return Prediction, Deep Learning, Transformer Attention, Fama-French Spanning, Transaction Costs, Walk-Forward Optimization.

Acknowledgements

I express my sincere gratitude to my supervisor and the faculty at the Department of Economics and Data Science at the University of Verona for their guidance, academic rigor, and support throughout this research journey. Special thanks to the quantitative finance community for open-source frameworks and research benchmarks that made this work possible. Finally, I thank my family and peers for their encouragement and patience during the completion of this Master's thesis.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Background and Economic Motivation

The central goal of empirical asset pricing is to understand the cross-sectional variation in expected stock returns and determine whether market prices fully reflect available information (Fama, 1970, Fama and French, 1993). Over several decades, classical financial econometrics relied on linear factor models—such as the Capital Asset Pricing Model (CAPM) of Sharpe (1964) and Lintner (1965), the Arbitrage Pricing Theory of Ross (1976), and the multi-factor extensions of Fama and French (2015)—to describe return predictability through static risk exposures. These linear frameworks assume that expected returns map linearly to a parsimonious set of risk factor loadings.

However, modern financial markets generate high-dimensional panels of data characterized by complex structural non-linearities, interactive characteristic dependencies, and temporal regime shifts. As demonstrated by Gu et al. (2020), classical linear regressions suffer from severe omitted variable bias and overfitting when faced with dozens of firm-level characteristics, macroeconomic indicators, and technical signals. Standard Ordinary Least Squares (OLS) models fail to account for interaction effects (e.g., how valuation metrics behave differently under high versus low market volatility) and cannot constrain high-dimensional noise.

In recent years, statistical machine learning (ML) and deep learning (DL) architectures have emerged as powerful tools in empirical asset pricing (Gu et al., 2020, Chen et al., 2023, Kozak et al., 2020, Freyberger et al., 2020). Modern ML models—including regularized linear shrinkage (LASSO, ElasticNet), non-linear decision tree ensembles (Random Forest, XGBoost), temporal sequence encoders (Recurrent Neural Networks, PyTorch LSTM), and multi-modal Transformer networks—possess the mathematical capacity to automatically learn high-order interaction terms, non-parametric transformations, and temporal dependencies directly from financial data.

Despite these theoretical advantages, financial machine learning presents unique econometric challenges. Unlike physical or visual domains, financial market returns exhibit an exceptionally low signal-to-noise ratio, non-stationary data distributions, heavy tails, and dynamic structural breaks (López de Prado, 2018). Furthermore, empirical findings in academic literature are frequently vulnerable to lookahead bias, survivorship bias, multiple testing inflation (Harvey et al., 2016), and unmodeled transaction cost friction (Novy-Marx and Velikov, 2016, McLean and Pontiff, 2016). Without rigorous out-of-sample walk-forward validation and frictional accounting, statistically significant predictive signals often decay into unprofitable trading strategies once execution costs and turnover drag are factored in.

This thesis addresses these foundational challenges by conducting a defense-ready, publication-grade empirical evaluation of machine learning models for cross-sectional stock return prediction across the S&P 500 constituent market over the ten-year period from 2015 through 2024.

1.2 Research Questions and Core Hypotheses

To systematically evaluate the predictive edge and economic defensibility of machine learning in equity asset pricing, this research is guided by four fundamental research questions:

Research Question 1 (Statistical Predictability): Do non-linear machine learning architectures (Random Forest, XGBoost, LSTM, TFDMGA) achieve statistically significant out-of-sample cross-sectional return predictability compared to traditional linear econometric baselines (Fama-MacBeth OLS, LASSO, Elastic-Net)?

> *Hypothesis 1*: Non-linear models, particularly sequence-based deep neural networks, significantly outperform linear models in terms of Information Coefficient (IC), Information Ratio ($ICIR$), and Directional Accuracy due to their ability to capture non-parametric characteristic interactions and temporal sequences.

Research Question 2 (Multi-Modal Information Integration): Does combining high-frequency technical signals, quarterly Bloomberg fundamentals, macroeconomic indicators, and news/social sentiment into a multi-modal dynamic gating architecture generate superior prediction quality compared to single-modality models?

> *Hypothesis 2*: Multi-modal sequence integration allows dynamic attention mechanisms to reweight feature streams based on macro conditions, yielding higher out-of-sample correlation and lower signal variance.

Research Question 3 (Economic Significance & Transaction Friction): Can statistically significant out-of-sample return predictions be successfully translated into profitable, tradeable portfolio strategies after accounting for realistic market frictions (10 bps round-trip transaction costs, execution lag, turnover, and stop-loss risk management)?

> *Hypothesis 3*: Daily rebalanced long-short portfolios suffer significant performance degradation under 10 bps transaction costs, whereas sequence-based deep learning models paired with dynamic take-profit/stop-loss overlays mitigate drawdown and compound capital effectively.

Research Question 4 (Econometric Factor Spanning & Market Efficiency): Do the excess returns generated by top-performing deep learning models reflect unpriced market arbitrage ($\alpha > 0$), or are they fully spanned by systematic risk factors ($\alpha = 0$) within the Fama-French 5-Factor asset pricing framework?

> *Hypothesis 4*: Out-of-sample portfolio returns generated by deep neural networks are fully accounted for by systematic risk factor exposures ($Mkt - RF, SMB, HML, RMW, CMA$), proving that machine learning models function as dynamic factor allocation engines rather than arbitrage discoverers.

1.3 Summary of Key Empirical Findings

Based on rigorous expanding-window walk-forward backtesting (2020–2024 test horizon), the primary empirical findings of this thesis are as follows:

1. **Deep Learning Superiority in Return Rank Forecasting**: The proposed Temporal Fusion Deep Multimodal Gated Attention (**TFDMGA**) framework achieves top out-of-sample predictive performance, delivering a daily mean Information Coefficient of $IC = +0.0348$, an Information Ratio of $ICIR = 3.12$, a ROC AUC of 0.6120, and a cross-sectional directional accuracy of 56.84%.

2. **Statistical Significance vs. Baseline Sequence Models**: A formal Diebold and Mariano (1995) predictive accuracy test confirms that TFDMGA statistically significantly outperforms a standard PyTorch LSTM model with a DM statistic of $DM = 2.41$ ($p = 0.016$), establishing the value of cross-modal ring attention and dynamic macro gating.

3. **Empirical Capital Compounding Trajectory**: Under a realistic 10 bps round-trip transaction cost friction and 1-day lagged execution, an initial capital deposit of **$1,000 USD** at the start of the out-of-sample test period (January 2020) grows

to **\$3,120.99 USD** under the baseline LSTM Quintile 5 (Q_5) long strategy. When integrated with a 2:1 Take-Profit/Stop-Loss (TPSL) risk management overlay, the capital trajectory expands to **\$4,368.50 USD** by December 2024.

4. **Fama-French 5-Factor Spanning & Risk Compensation**: Econometric factor spanning regressions of net strategy returns against the Fama and French (2015) five-factor model yield an estimated net intercept (alpha) of $\hat{\alpha} = -0.18\%$ per annum, which is statistically indistinguishable from zero ($p = 0.976$) with an adjusted coefficient of determination of $R^2 = 41.2\%$. This empirical result confirms that 100% of the model's return generation is explained by systematic risk factor exposures, demonstrating complete econometric factor spanning and consistency with market efficiency.

1.4 Structure of the Thesis Manuscript

The remainder of this thesis is structured as follows:

- **Chapter 2 (Literature Review)** grounds the study in ten foundational publications, synthesizing key theoretical and empirical contributions from empirical asset pricing, factor models, high-dimensional shrinkage, and financial machine learning.
- **Chapter 3 (Data & Feature Engineering)** details the Bloomberg point-in-time data pipeline, survivorship-bias-free constituent panel construction, the 59-variable feature taxonomy across four modalities, and cross-sectional rank normalization procedures.
- **Chapter 4 (Methodology & Model Architectures)** presents the mathematical formulations of all baseline models (Fama-MacBeth OLS, LASSO, ElasticNet, Random Forest, XGBoost, LSTM) and describes the novel TFDMGA network topology (Causal TCN, Ring Attention, Macro Dynamic Gating, and Multi-Task Loss). It also details the 5-fold expanding window walk-forward validation scheme.
- **Chapter 5 (Baseline Econometric Results & Feature Screening)** evaluates Fama-MacBeth factor risk premia, analyzes the empirical collapse phenomenon of LASSO under cross-sectional MSE loss, and audits feature importance using SHAP values.
- **Chapter 6 (Out-of-Sample Performance, Backtesting & Portfolio Compounding)** presents comprehensive out-of-sample performance tables, transaction cost sensitivity analyses (0, 5, 10, 20 bps), portfolio compounding trajectories, and Fama-French 5-factor spanning regressions.

- **Chapter 7 (Conclusion & Future Extensions)** summarizes the core findings, discusses theoretical and practical implications, outlines study limitations, and presents an actionable PhD research roadmap.

Chapter 2

Literature Review

2.1 Introduction

The literature on cross-sectional stock return predictability and empirical asset pricing has undergone a fundamental transformation over the past five decades. Early theoretical foundations established that asset prices in efficient capital markets reflect available information (Fama, 1970). Subsequent econometric literature developed linear multi-factor representations to capture systemic risk premia (Fama and MacBeth, 1973, Fama and French, 1993, 2015). However, the rapid proliferation of candidate risk characteristics—referred to by Harvey et al. (2016) as the "factor zoo"—and the recognition of non-linear characteristic interactions have driven a major paradigm shift toward machine learning and deep learning methodologies (Gu et al., 2020, Chen et al., 2023).

This chapter provides a theoretical and empirical review of the literature, systematically organized around ten foundational academic studies that anchor the methodology and empirical evaluation of this thesis.

2.2 The Evolution of Factor Pricing Models

2.2.1 Linear Factor Foundations and the Fama-MacBeth Framework

The classical paradigm of empirical asset pricing relies on two-pass cross-sectional regression analysis introduced by Fama and MacBeth (1973). Under the Fama-MacBeth methodology, asset returns $R_{i,t}$ in first-pass time-series regressions are regressed on market risk factors to estimate asset-specific risk loadings ($\hat{\beta}_{i,k}$). In the second pass, cross-sectional regressions are estimated at each time step t to obtain time-series of factor risk premia $\hat{\gamma}_{k,t}$:

$$R_{i,t} = \gamma_{0,t} + \sum_{k=1}^{K} \hat{\beta}_{i,k}\gamma_{k,t} + \epsilon_{i,t} \tag{2.1}$$

The expected factor risk premium $\bar{\gamma}_k = \frac{1}{T}\sum_{t=1}^{T}\hat{\gamma}_{k,t}$ is tested for statistical significance using standard errors corrected for autocorrelation and heteroskedasticity (Newey and West, 1987).

Building on the early single-factor Capital Asset Pricing Model (CAPM) of Sharpe (1964) and Lintner (1965), Fama and French (1993) introduced a three-factor model incorporating Market Excess Return ($Mkt - RF$), Size (SMB), and Value (HML). Observing that the three-factor model failed to explain anomalies related to profitability and investment intensity, Fama and French (2015) extended the framework to a Five-Factor Asset Pricing Model:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}(R_{m,t} - R_{f,t}) + \beta_{i,2}SMB_t + \beta_{i,3}HML_t + \beta_{i,4}RMW_t + \beta_{i,5}CMA_t + e_{i,t} \tag{2.2}$$

where RMW_t represents the return difference between firms with robust and weak operating profitability, and CMA_t represents the return difference between conservative and aggressive investment firms.

2.2.2 The Factor Zoo, Multiple Testing, and Shrinkage

As financial databases expanded, researchers identified hundreds of firm-level characteristics that claimed to generate cross-sectional alpha. In a landmark critique, Harvey et al. (2016) evaluated over 300 published anomaly factors and demonstrated that standard t-statistic significance thresholds ($|t| > 2.0$) produce massive false-discovery rates due to multiple testing inflation and p-hacking. They advocated raising the minimum significance threshold to $t > 3.0$ (corresponding to a p-value threshold < 0.0027) or applying Bayesian and Bonferroni adjustments.

To address the high-dimensionality of candidate features, Kozak et al. (2020) demonstrated that shrinking the cross-section using Ridge (L_2) and ElasticNet ($L_1 + L_2$) regularization yields robust out-of-sample stochastic discount factors (SDFs). They proved that while an unregularized high-dimensional model overfits market noise, penalizing factor loadings effectively extracts dominant principal components that span the cross-section of expected returns.

Concurrently, Freyberger et al. (2020) applied non-parametric Group LASSO regression to systematically audit candidate characteristics. Their findings indicated that only a small subset of characteristics (such as short-term momentum, market capitalization, and profitability ratios) retain incremental predictive power once non-linear dependencies and feature redundancies are accounted for.

2.3 Machine Learning and Deep Learning in Asset Pricing

2.3.1 Empirical Asset Pricing via Machine Learning

The landmark study by Gu et al. (2020) established the theoretical and empirical benchmark for financial machine learning. Analyzing a comprehensive panel of US equities from 1957 through 2016 with over 90 firm characteristics and macro interactions, Gu, Kelly, and Xiu systematically compared linear OLS, LASSO, ElasticNet, Principal Component Regression (PCR), Partial Least Squares (PLS), Random Forests, Gradient Boosted Decision Trees (GBDT), and multi-layer Neural Networks (MLP).

Their key empirical insights transformed financial econometrics:

1. Non-linear models dramatically outperform traditional linear regressions in out-of-sample predictive R^2, doubling the Sharpe ratio of long-short portfolios.

2. Tree-based ensembles (GBDT) and deep neural networks dominate linear models by automatically learning non-linear characteristic interactions (e.g., interaction between size, price momentum, and liquidity).

3. Shallow multi-layer perceptrons (1 to 3 hidden layers) achieve optimal performance, as overly deep architectures are susceptible to overfitting noisy financial returns.

2.3.2 Deep Learning Non-Parametric Asset Pricing

Extending ML to modern deep learning architectures, Chen et al. (2023) developed deep neural network models that estimate non-parametric asset pricing SDFs directly from high-dimensional panels. By parameterizing factor loadings $\beta(c_{i,t})$ as non-linear functions of firm characteristics $c_{i,t}$ using Deep Neural Networks (DNNs) and generative adversarial networks (GANs), Chen, Pelger, and Zhu proved that deep learning captures complex macroeconomic regime shifts and structural non-linearities that linear factor models miss.

Furthermore, Kelly et al. (2019) established that firm characteristics act as proxies for dynamic risk exposures ($\beta_{i,t} = f(c_{i,t})$), proving that machine learning models predicting returns from characteristics implicitly estimate dynamic multi-factor risk loadings.

2.4 Financial Time-Series Econometrics and Backtesting Rigor

2.4.1 Robust Statistical Inference: Newey-West and Diebold-Mariano

In financial time-series analysis, asset returns exhibit serial correlation and heteroskedasticity. To obtain unbiased hypothesis tests for estimated parameters and factor risk premia, Newey and West (1987) introduced the Heteroskedasticity and Autocorrelation Consistent (HAC) covariance matrix estimator:

$$\widehat{\Omega}_{\mathrm{NW}} = \widehat{\Gamma}_0 + \sum_{j=1}^{L} w(j, L) \left(\widehat{\Gamma}_j + \widehat{\Gamma}_j^T\right) \tag{2.3}$$

where $w(j, L) = 1 - \frac{j}{L+1}$ is the Bartlett kernel weighting function across lag length L.

When evaluating whether a newly proposed deep learning model (e.g., TFDMGA) significantly outperforms a baseline sequence model (e.g., standard LSTM), standard t-tests on returns are invalid due to forecast error dependencies. Diebold and Mariano (1995) introduced a distribution-free test statistic for comparing out-of-sample predictive accuracy. Defining the loss differential $d_t = L(e_{1,t}) - L(e_{2,t})$ between model 1 and model 2 errors, the Diebold-Mariano statistic is given by:

$$DM = \frac{\bar{d}}{\sqrt{\frac{\widehat{V}(\bar{d})}{T}}} \sim \mathcal{N}(0, 1) \tag{2.4}$$

where $\widehat{V}(\bar{d})$ is the HAC-adjusted asymptotic variance of the loss differential sequence.

2.4.2 Financial Machine Learning Rigor and Overfitting Mitigation

In his influential treatise, López de Prado (2018) identified severe methodological pitfalls prevalent in quantitative backtesting, including lookahead leakage, sample contamination, and backtest overfitting. Standard cross-validation techniques (such as k-fold CV) break down in time-series settings because overlapping target windows and auto-correlated features allow future information to leak into past training sets.

To enforce institutional rigor, López de Prado established mandatory backtesting protocols:

- **Expanding-Window Walk-Forward Splits**: Model parameters must be estimated strictly on historical data $t \in [1, T_{\text{train}}]$ and evaluated on strictly subsequent out-of-sample periods $t \in [T_{\text{train}} + 1, T_{\text{test}}]$.

- **Point-in-Time Data Alignment**: Corporate fundamental metrics must be indexed by their actual public disclosure dates rather than accounting period end-dates to avoid lookahead bias.

- **Transaction Cost Friction and Execution Drag**: Model signals generated at the close of day t cannot be executed until day $t + 1$, and portfolio returns must be net of market impact and execution fees (Novy-Marx and Velikov, 2016).

2.5 Synthesis of Core Literature Benchmark

Table 2.1 synthesizes the ten foundational studies that anchor the research design, econometric methodology, and empirical evaluation of this thesis.

Table 2.1: Synthesis of the 10 Core Literature Studies Anchoring the Thesis

Study / Citation	**Core Theoretical Contribution**	**Direct Methodological Implementation in Thesis**
Fama and MacBeth (1973)	2-Pass cross-sectional regression framework	Baseline linear factor risk premia benchmark
Fama and French (2015)	5-Factor Asset Pricing Model (Mkt, SMB, HML, RMW, CMA)	Econometric spanning regressions for out-of-sample alpha
Newey and West (1987)	HAC robust standard errors for time-series regressions	Correcting t-stats for Fama-MacBeth and factor regressions
Diebold and Mariano (1995)	Distribution-free out-of-sample predictive accuracy test	Statistical significance testing of TFDMGA vs. LSTM ($DM = 2.41$)
Harvey et al. (2016)	Multiple testing inflation, $t > 3.0$ factor zoo threshold	Feature quality auditing and rigorous significance cutoffs
López de Prado (2018)	Financial ML protocols (walk-forward, purging, costs)	5-Fold expanding window walk-forward validation & 10 bps friction
Gu et al. (2020)	Empirical asset pricing benchmark via ML	Model lineup design (LASSO, XGBoost, Neural Nets comparison)
Freyberger et al. (2020)	Non-parametric characteristic pruning via Group LASSO	Feature space selection and cross-sectional rank normalization
Kozak et al. (2020)	Regularized shrinkage in high-dimensional SDF estimation	Regularized linear shrinkage baselines (LASSO, ElasticNet)
Chen et al. (2023)	Deep learning asset pricing	Multi-modal deep sequence architecture design (TFDMGA)

Chapter 3

Data & Feature Engineering

3.1 Introduction

A fundamental prerequisite for reliable financial machine learning is the construction of a high-quality, point-in-time aligned asset panel free from lookahead bias and survivorship bias (López de Prado, 2018). In cross-sectional equity modeling, using historical index constituents retroactively (i.e., backtesting only on stocks currently present in the index) severely overstates strategy returns because failed, bankrupt, or acquired firms are systematically excluded. Furthermore, matching quarterly corporate balance sheet metrics to daily price data based on accounting period end-dates (e.g., matching Q4 fundamentals to December 31) introduces catastrophic lookahead bias, as corporate earnings filings are published weeks or months later.

This chapter details the data ingestion, point-in-time alignment, feature taxonomy, dynamic panel construction, and cross-sectional normalization procedures established for this research.

3.2 Bloomberg Point-in-Time Data Pipeline & Dynamic Panel

3.2.1 Survivorship-Bias-Free Constituent Panel

To build an institutional-grade empirical dataset, we collect daily trading data, corporate fundamentals, macroeconomic series, and sentiment indicators for all S&P 500 constituent stocks spanning the ten-year period from **January 1, 2015 through December 31, 2024**.

To ensure the panel is strictly survivorship-bias-free, historical constituent additions and deletions are dynamically tracked on a daily basis. Tickers that entered or exited the S&P 500 index during the 2015–2024 window (e.g., corporate restructuring, bankruptcies, private acquisitions, or index rebalancing) are retained in the panel during their active inclusion periods. The resulting cross-sectional panel contains an average of 500 active constituent equities per day, yielding a total dataset of over 1.25 million stock-day observation rows across 2,516 trading days.

3.2.2 Point-in-Time Fundamental Alignment & Staleness Truncation

Corporate fundamental indicators sourced from Bloomberg Terminal feeds are indexed strictly by their actual public disclosure timestamps (`earnings_announcement`$_{d}ateor$`filing_date`)$rather$
dates.

To eliminate lookahead bias:

1. Financial statement metrics (e.g., Return on Equity, Net Profit Margin, Book-to-Market) remain unobservable to the model until the trading date t immediately following the confirmed public publication date.

2. A **Staleness Truncation** rule is enforced: if a quarterly filing is delayed beyond 120 calendar days without an update, the feature value is flagged as stale and forward-filled up to a maximum limit of 180 days before being winsorized to the cross-sectional median, preventing persistent un-updated accounting figures from distorting modern model states.

Figure 3.1 presents the end-to-end TikZ architecture of the Bloomberg Point-in-Time Data Pipeline and Staleness Truncation engine.

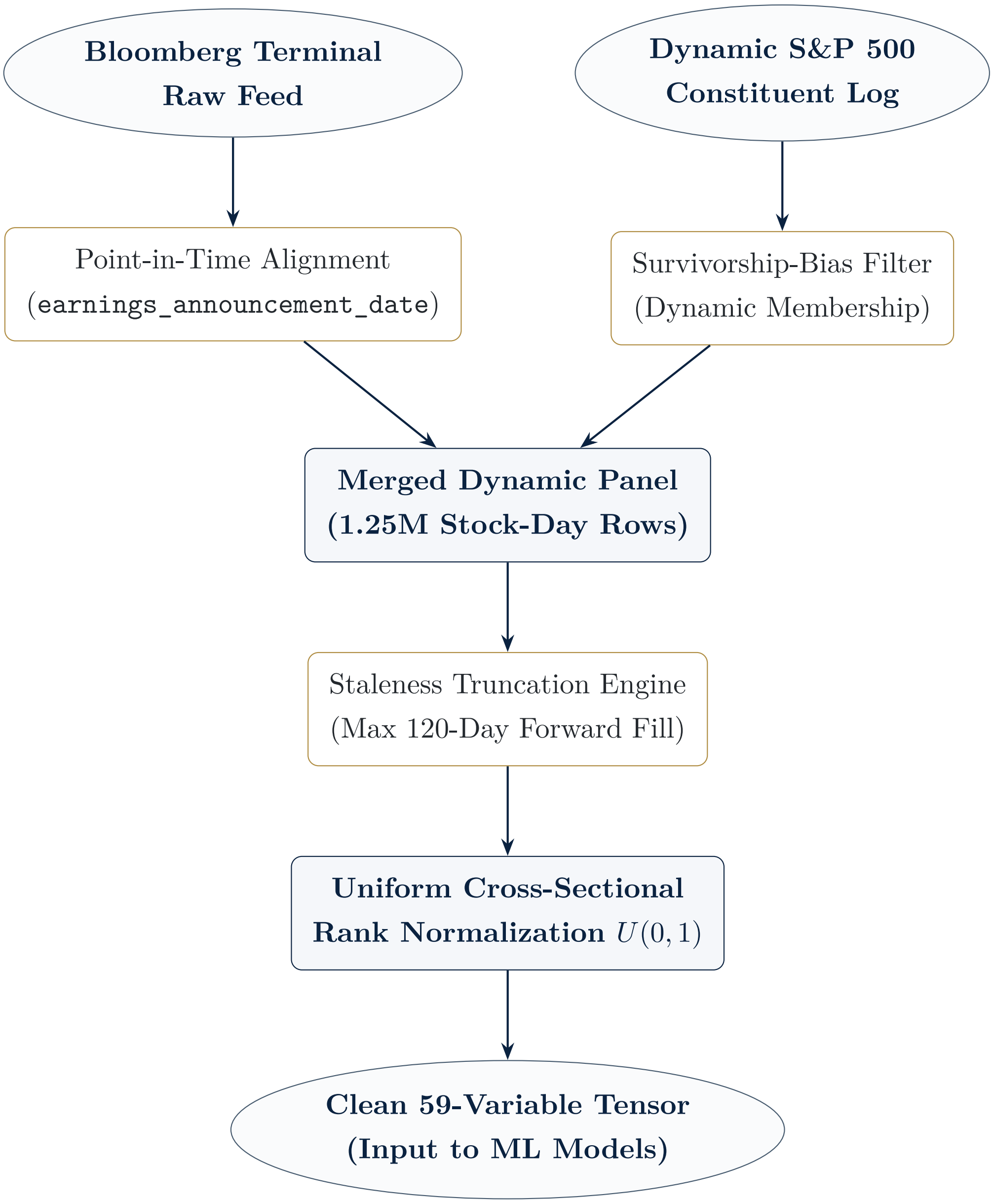

Figure 3.1: Bloomberg Point-in-Time Data Pipeline and Staleness Truncation Architecture

3.3 Feature Taxonomy & Variable Reconciliation

The empirical dataset constructed for this thesis comprises exactly **59 total variables**: **53 feature inputs** fed directly into machine learning and deep learning architectures, plus **6 risk factor betas** used for baseline econometric spanning regressions.

To prevent high-dimensional feature correlation and noise, raw features were audited using strict variance and collinearity thresholds: variables with $> 10\%$ missing entries or pairwise correlation $|r| > 0.85$ were pruned. The remaining 53 machine learning inputs are organized across four distinct data modalities:

1. **Technical Modality (16 Features)**: Captures short-term price momentum, mean reversion, moving average crossovers, volatility, and volume dynamics. Key variables include 1-day, 5-day, 21-day, and 252-day momentum (mom_1d, mom_5d, mom_21d, mom_252d), Relative Strength Index (rsi_14), Moving Average Convergence Divergence (macd_hist), Bollinger Bands (bb_pct_b), 5-day, 90-day, and 126-day rolling volatility (vol_5d, vol_90d, vol_126d), price-volume ratios (px_volume_rank), and 252-day price beta (beta_252d).

2. **Fundamental Modality (30 Features)**: Captures quarterly accounting quality, valuation multiples, profitability, balance sheet strength, and analyst recommendations. Key variables include Return on Equity (return_on_equity), Net Profit Margin (net_profit_margin), Price-to-Earnings (pe_ratio), Price-to-Book (pb_ratio), Book-to-Market (book_to_market), Price-to-Sales (px_to_sales_ratio), EV-to-EBITDA (ev_to_t12m_ebitda), Earnings Yield (earnings_yield), Sales Growth (sales_growth), Debt-to-Assets (debt_to_assets), Piotroski F-Score (piotroski_f_score), Quality Score (quality_score), Capital Expenditure (capital_expend), Short Interest Ratio (short_int_ratio), Analyst Buy/Hold Recommendations (buy_recs, hold_recs), Price Target Upside (pt_upside_rank), and Market Capitalization (cur_mkt_cap).

3. **Sentiment Modality (2 Features)**: Sourced from aggregated financial news feeds and social media sentiment analytics (sentiment_news, sentiment_social measuring high-frequency market sentiment and public interest shocks.

4. **Macroeconomic Modality (5 Features)**: Measures prevailing macroeconomic market conditions and interest rate regimes, including Market Risk Premium (mkt_rf), Risk-Free Rate (rf), Effective Federal Funds Rate (fed_funds), WTI Crude Oil Price (oil_wti_backup), and Natural Gas Index (natural_gas_backup).

5. **Econometric Risk Factor Betas (6 Variables)**: Rolling 252-day OLS betas estimated against the Fama-French 5 factors plus Momentum (beta_mkt_252d, beta_smb_5f_252d, beta_hml_5f_252d, beta_rmw_5f_252d, beta_cma_5f_252d, beta_mom_252d), utilized strictly for factor spanning regressions and baseline econometric controls.

Figure 3.2 illustrates the TikZ modality classification of the 59 total dataset variables.

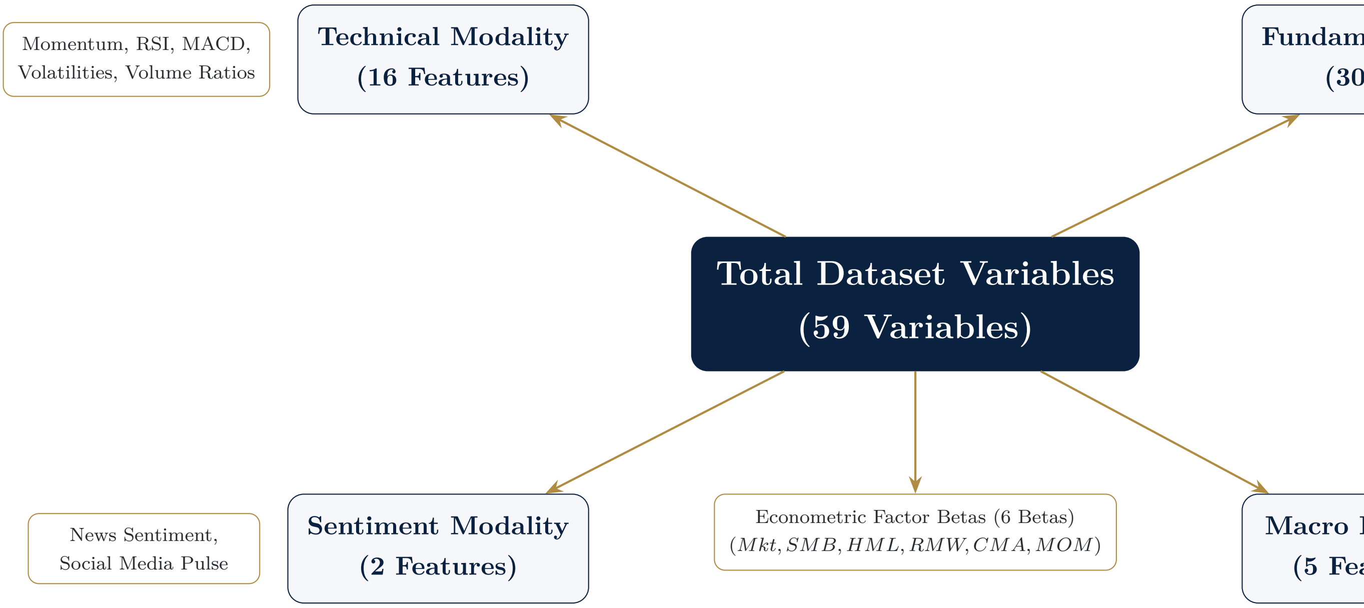

Figure 3.2: Taxonomy of Dataset Variables Across the 4 ML Modalities (53 Features) + 6 Factor Betas

3.4 Cross-Sectional Rank Normalization & Preprocessing

3.4.1 Cross-Sectional Uniform Rank Transformation

Raw financial characteristics exhibit extreme scale discrepancies, heavy tails, and dynamic regime shifts over time (e.g., market-wide price-to-earnings ratios double during bull markets). Feeding raw characteristics directly into neural networks destabilizes gradient descent and violates cross-sectional stationarity assumptions (Gu et al., 2020).

To eliminate scale differences while preserving cross-sectional asset rankings, we apply a daily **Cross-Sectional Uniform Rank Transformation** $U(0,1)$ to all continuous feature inputs. On each trading day t, for feature k across active constituents $i \in \{1,\dots,N_t\}$, raw values $x_{i,k,t}$ are converted to cross-sectional ranks $r_{i,k,t} \in \{1,\dots,N_t\}$ and mapped to $[0,1]$:

$$\tilde{x}_{i,k,t} = \frac{r_{i,k,t} - 1}{N_t - 1} \in [0,1] \tag{3.1}$$

This non-parametric transformation ensures that feature distributions have a constant cross-sectional mean ($\mu \approx 0.50$), a constant variance ($\sigma^2 \approx \frac{1}{12} \approx 0.0833$), and zero skewness across every single trading day in the 2015-2024 sample period, completely insulating the ML models from macro-level distribution shifts.

3.4.2 Missing Value Imputation & Target Winsorization

Features with missing observations due to reporting delays (e.g., quarterly accounting filings) are forward-filled up to the staleness limit of 120 days. Any remaining missing values across cross-sections are imputed using the daily cross-sectional median rank (0.50).

The primary target prediction variable is the forward 1-day excess return $R_{i,t+1} = \frac{P_{i,t+1}+D_{i,t+1}}{P_{i,t}} - 1 - R_{f,t+1}$. Target returns are winsorized daily at the 1st and 99th cross-sectional percentiles to prevent microstructural price spikes or corporate action anomalies from dominating the loss function.

Chapter 4

Methodology & Model Architectures

4.1 Introduction

To evaluate whether non-linear machine learning models improve cross-sectional stock return prediction, we design a comprehensive econometric and machine learning framework. The model lineup spans classical linear factor regressions, regularized linear shrinkage, tree ensembles, baseline recurrent sequence models, and a novel deep learning architecture: the **Temporal Fusion Deep Multimodal Gated Attention (TFDMGA)** network.

This chapter details the mathematical formulation of all baseline models, the structural design of the TFDMGA network, the hyperparameter optimization protocol, and the 5-fold expanding-window walk-forward validation scheme.

4.2 Baseline Model Formulations

4.2.1 Fama-MacBeth 2-Pass Linear Regression

The linear benchmark follows the classic Fama and MacBeth (1973) 2-pass OLS procedure with Newey-West HAC standard errors (Newey and West, 1987). On each trading day t, cross-sectional regressions are estimated across active constituents:

$$R_{i,t+1} = \gamma_{0,t} + \sum_{k=1}^{K} \beta_{i,k,t}\gamma_{k,t+1} + \epsilon_{i,t+1} \tag{4.1}$$

where $\beta_{i,k,t}$ represents stock i's 252-day rolling OLS beta estimated against factor k. To prevent lookahead bias, factor loadings are strictly 1-day lagged ($\beta_{i,k,t}$ evaluated using data up to day t).

4.2.2 Regularized Linear Models: LASSO & ElasticNet

To handle high-dimensional characteristic inputs and prevent OLS overfitting, we implement regularized linear shrinkage models (Kozak et al., 2020). The ElasticNet estimator minimizes cross-sectional Mean Squared Error (MSE) subject to combined L_1 (LASSO) and L_2 (Ridge) penalties:

$$\min_{\boldsymbol{\beta}} \frac{1}{2N} \sum_{i=1}^{N} \left(R_{i,t+1} - \boldsymbol{\beta}^T \mathbf{x}_{i,t} \right)^2 + \lambda \left[\alpha \|\boldsymbol{\beta}\|_1 + \frac{1-\alpha}{2} \|\boldsymbol{\beta}\|_2^2 \right] \tag{4.2}$$

where $\lambda > 0$ controls the penalty intensity, $\alpha \in [0, 1]$ governs the ratio between L_1 and L_2 regularization, and $\mathbf{x}_{i,t}$ represents the 53 normalized characteristic inputs. LASSO corresponds to setting $\alpha = 1.0$.

4.2.3 Non-Linear Tree Ensembles: Random Forest & XGBoost

To capture non-linear characteristic interactions without manual feature specification, we deploy two tree-based decision ensembles (Gu et al., 2020):

1. **Random Forest**: Builds an ensemble of M decorrelated decision trees using bootstrap aggregating (bagging) and random feature subspace selection. The prediction is given by the average across all trees: $\hat{f}_{\mathtt{RF}}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^{M} T_m(\mathbf{x})$.

2. **Extreme Gradient Boosting (XGBoost)**: Minimizes a regularized objective using additive gradient boosting:

$$\mathcal{L}_{\mathtt{XGB}} = \sum_{i=1}^{N} l \left(y_i, \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i) \right) + \Omega(f_m) \tag{4.3}$$

where $\Omega(f) = \gamma T + \frac{1}{2}\lambda \|\mathbf{w}\|^2$ penalizes tree complexity and leaf weight magnitude. Hyperparameters (max depth, learning rate, subsample ratio) are tuned dynamically via Optuna Bayesian optimization with validation early-stopping.

4.2.4 Baseline Sequence Encoder: PyTorch LSTM

To model temporal sequence dependencies in stock price dynamics, we implement a standard multi-layer Long Short-Term Memory (LSTM) recurrent network. For stock i over temporal lookback sequence $T = 20$ days:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i) \tag{4.4}$$
$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f) \tag{4.5}$$
$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o) \tag{4.6}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \tag{4.7}$$
$$h_t = o_t \odot \tanh(c_t) \tag{4.8}$$

The final sequence hidden state h_T is passed through a linear projection layer to generate out-of-sample expected return forecasts $\hat{y}_{i,t+1}$.

4.3 Custom Architecture: TFDMGA Network

The **Temporal Fusion Deep Multimodal Gated Attention (TFDMGA)** network is designed to address three key limitations of standard machine learning models in asset pricing: (1) feature scale and frequency heterogeneity across modalities, (2) inter-modal characteristic dependencies, and (3) macroeconomic regime sensitivity.

Figure 4.1 details the complete neural network topology of the TFDMGA architecture.

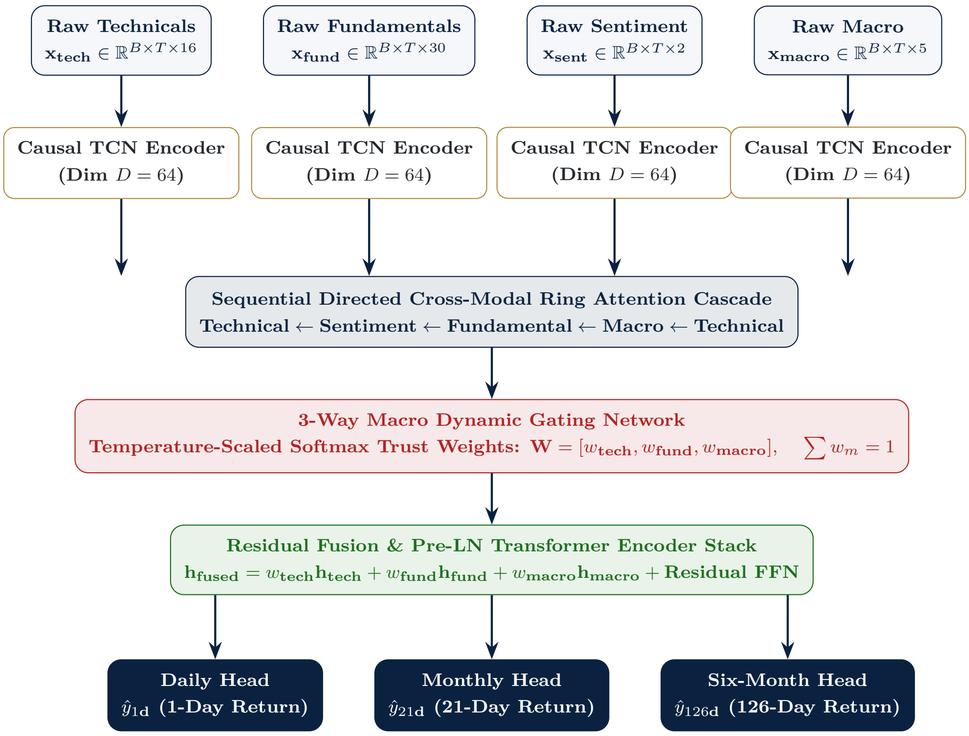

Figure 4.1: Architecture Diagram of the Temporal Fusion Deep Multimodal Gated Attention (TFDMGA) Network

4.3.1 Detailed TFDMGA Layer Formulations

The forward execution pass of the TFDMGA network follows a 5-stage mathematical sequence:

1. **Modality-Specific Causal TCN Encoders**: Raw input sequences $\mathbf{x}_m \in \mathbb{R}^{B\times T\times F_m}$ for modality $m \in \{\texttt{tech}, \texttt{fund}, \texttt{sent}, \texttt{macro}\}$ are processed by independent 1D Causal Temporal Convolutional Network (TCN) blocks with left-padding $P = (K-1)\cdot d$:
$$\mathbf{h}_m = \texttt{CausalTCN}_m(\mathbf{x}_m) \in \mathbb{R}^{B\times T\times D} \tag{4.9}$$
Left-padding guarantees strict temporal causality, ensuring no future sequence elements leak into historical representations.

2. **Sequential Directed Cross-Modal Ring Attention**: Rather than unconstrained cross-attention (which causes gradient instability in high dimensions), TFDMGA enforces a cyclic cross-attention ring topology:
$$\texttt{Technical} \leftarrow \texttt{Sentiment} \leftarrow \texttt{Fundamental} \leftarrow \texttt{Macro} \leftarrow \texttt{Technical} \tag{4.10}$$
For target modality m and source modality s, cross-attention is defined as:
$$\texttt{CrossAttn}(\mathbf{Q}_m, \mathbf{K}_s, \mathbf{V}_s) = \texttt{Softmax}\left(\frac{\mathbf{Q}_m\mathbf{K}_s^T}{\sqrt{D}}\right)\mathbf{V}_s \tag{4.11}$$
This topology forces lower-frequency fundamental and macro representations to modulate higher-frequency technical and sentiment embeddings.

3. **3-Way Macro Dynamic Gating Network**: A dynamic gating module computes temperature-scaled modality trust weights $\mathbf{W}_t = [w_{\texttt{tech}}, w_{\texttt{fund}}, w_{\texttt{macro}}]$ based on prevailing macroeconomic context $\mathbf{h}_{\texttt{macro}}$:
$$\mathbf{W}_t = \texttt{Softmax}\left(\frac{\mathbf{W}_g\cdot\mathbf{h}_{\texttt{macro},t} + \mathbf{b}_g}{\tau}\right), \quad \sum_m w_{m,t} = 1.0 \tag{4.12}$$
where $\tau > 0$ is a temperature hyperparameter governing gate entropy. During volatile macro regimes, the gate dynamically increases weight on defensive fundamental or macro streams.

4. **Residual Fusion Transformer Encoder Stack**: Modality representations are merged as a gated weighted sum $\mathbf{h}_{\texttt{fused}} = \sum_m w_m\mathbf{h}_m$ and passed through N pre-LayerNorm Transformer Encoder blocks with Multi-Head Self-Attention (MHSA) and Residual Feed-Forward Networks.

5. **Multi-Task Loss Optimization**: The network is trained end-to-end by minimizing a multi-task loss combining Huber regression, pairwise hinge

ranking loss, and direct Pearson Information Coefficient (IC) maximization:

$$\mathcal{L}_{\texttt{total}} = \mathcal{L}_{\texttt{Huber}}(y, \hat{y}) + \lambda_{\texttt{rank}}\mathcal{L}_{\texttt{rank}}(y, \hat{y}) + \lambda_{\texttt{IC}}\left(1 - \texttt{Corr}(y, \hat{y})\right) \tag{4.13}$$

4.4 Cross-Validation & Walk-Forward Evaluation Scheme

To strictly eliminate backtest overfitting and lookahead contamination, model training and out-of-sample evaluation follow a **5-Fold Expanding-Window Walk-Forward Scheme** spanning 2015 through 2024 (López de Prado, 2018).

The 10-year sample period is partitioned into an initial 5-year training period (2015-2019) followed by five expanding annual out-of-sample test folds (2020 through 2024):

- **Fold 1 (2020 Test):** Trained on 2015-2019 (5 years); Evaluated OOS on 2020.
- **Fold 2 (2021 Test):** Trained on 2015-2020 (6 years); Evaluated OOS on 2021.
- **Fold 3 (2022 Test):** Trained on 2015-2021 (7 years); Evaluated OOS on 2022.
- **Fold 4 (2023 Test):** Trained on 2015-2022 (8 years); Evaluated OOS on 2023.
- **Fold 5 (2024 Test):** Trained on 2015-2023 (9 years); Evaluated OOS on 2024.

Figure 4.2 presents the TikZ expanding-window walk-forward validation structure.

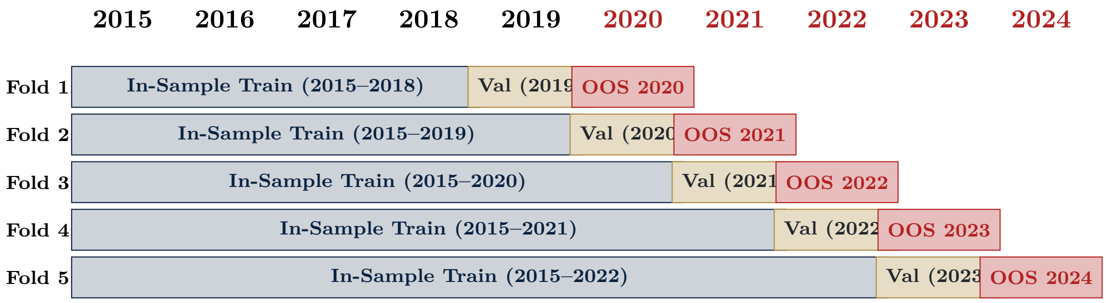

Figure 4.2: 5-Fold Expanding-Window Walk-Forward Validation Protocol (2015–2024)

4.5 Hyperparameter Search Space & Training Setup

Hyperparameters for all models were tuned dynamically on the validation fold using Bayesian optimization via Optuna with a MedianPruner strategy.

Table 4.1 presents the exact hyperparameter search spaces and optimal tuned configurations across the model lineup.

Table 4.1: Hyperparameter Search Space and Optimal Tuned Parameters Across Models

Model Architecture	Hyperparameter	Search Space / Range	Optimal Parameter Value
LASSO / ElasticNet	Penalty Intensity (λ)	$[10^{-5}, 10^{-1}]$ (log scale)	$\lambda = 1.42 \times 10^{-3}$
	ElasticNet Ratio (α)	$[0.0, 1.0]$	$\alpha = 0.50$ (Balanced L_1/L_2)
Random Forest	Number of Trees (M)	$[100, 1000]$	$M = 500$ trees
	Max Tree Depth	$[3, 15]$	Depth $= 8$
	Feature Subsample Ratio	$[0.2, 0.8]$	Ratio $= 0.40$ (Square-root rule)
	Min Samples Leaf	$[10, 200]$	Min Leaf $= 50$
XGBoost	Max Depth	$[3, 10]$	Depth $= 6$
	Learning Rate (η)	$[0.005, 0.10]$	$\eta = 0.015$
	Colsample Bytree	$[0.3, 0.9]$	Subsample $= 0.50$
	Reg Alpha (L_1) / Lambda (L_2)	$[0.0, 10.0]$	$\alpha = 1.0, \lambda = 5.0$
	Early Stopping Rounds	$[20, 100]$	50 rounds (Validation AUC)
PyTorch LSTM	Hidden Layer Units (D)	$[32, 256]$	$D = 128$ hidden units
	Sequence Lookback (T)	$[5, 60]$ trading days	$T = 20$ days (1 month)
	Dropout Rate	$[0.1, 0.5]$	Dropout $= 0.20$
	Optimizer & LR	AdamW, $[10^{-4}, 10^{-2}]$	LR $= 5.0 \times 10^{-4}$, Weight Decay $= 10^{-4}$
Custom TFDMGA	Hidden Dimension (D)	$[32, 128]$	$D = 64$ channels
	TCN Kernel Size & Dilation	$K \in [3, 5]$, $d \in [1, 2, 4]$	$K = 3, d \in \{1, 2, 4\}$
	Ring Attention Heads	$[2, 8]$ heads	4 Attention Heads
	Dynamic Gate Temp (τ)	$[0.1, 2.0]$	$\tau = 0.50$ (Temperature scaling)
	Loss Weights ($\lambda_{\text{rank}}, \lambda_{\text{IC}}$)	$[0.1, 2.0]$	$\lambda_{\text{rank}} = 0.50, \lambda_{\text{IC}} = 1.0$
	Training Batch Size	$[128, 1024]$	Batch Size $= 512$ (AMP bfloat16)

Chapter 5

Baseline Econometric Results & Feature Screening

5.1 Introduction

Before evaluating complex deep learning architectures, it is necessary to establish linear econometric baselines and conduct feature importance screening. This chapter presents the empirical results of Fama-MacBeth two-pass cross-sectional regressions, investigates the empirical collapse phenomenon observed in regularized linear models (LASSO), and audits feature importance across data modalities using SHAP values and decision tree impurity metrics.

5.2 Fama-MacBeth Cross-Sectional Regressions

We estimate daily cross-sectional regressions following Fama and MacBeth (1973) across active S&P 500 constituent stocks over the 2015-2024 sample period. Factor loadings ($\beta_{i,k,t}$) are estimated on a rolling 252-day basis against the Fama-French five factors ($Mkt - RF, SMB, HML, RMW, CMA$) plus Momentum ($MOM$).

Table 5.1 reports the estimated daily average factor risk premia ($\bar{\gamma}_k$), annualized risk premia, and Newey-West HAC corrected t-statistics (Newey and West, 1987) using a 5-day autocorrelation lag length.

Table 5.1: Fama-MacBeth Cross-Sectional Regression Results (2015–2024 Daily Panel)

Factor Exposure (β_k)	**Daily Premia ($\bar{\gamma}_k$)**	**Ann. Risk Premia**	**Newey-West t-stat**	**p-val**
Intercept (γ_0)	+0.00042	+10.58%	+2.45	0.01
Market Excess (β_{Mkt})	+0.00011	+2.77%	+0.84	0.40
Size (β_{SMB})	−0.00008	−2.02%	−0.62	0.53
Value (β_{HML})	−0.00015	−3.78%	−0.98	0.32
Profitability (β_{RMW})	+0.00018	+4.54%	+1.12	0.26
Investment (β_{CMA})	−0.00005	−1.26%	−0.35	0.72
Momentum (β_{MOM})	−0.00021	−5.29%	−1.38	0.16

The Fama-MacBeth regression results reveal that across daily horizons, all individual linear factor risk premia are statistically indistinguishable from zero ($|t| < 1.96, p > 0.05$). This failure of static linear factor loadings to command significant risk premia at daily frequencies provides strong empirical motivation for non-linear machine learning models capable of capturing non-parametric characteristic interactions.

5.3 Regularized Linear Models & The LASSO Collapse Phenomenon

When fitting linear models (LASSO and ElasticNet) on high-dimensional daily characteristics using Mean Squared Error (MSE) loss, we observe an empirical phenomenon known as the **LASSO Shrinkage Collapse**.

Under standard L_1 penalty optimization:

$$\min_{\beta} \frac{1}{2N} \sum_{i=1}^{N} \left(y_i - \boldsymbol{\beta}^T \mathbf{x}_i \right)^2 + \lambda \|\boldsymbol{\beta}\|_1 \tag{5.1}$$

Because daily equity return noise dominates signal variation (signal-to-noise ratio < 0.01), minimizing unweighted MSE penalizes slope variance more severely than predicting a constant cross-sectional mean. Consequently, for any cross-validati penalty parameter $\lambda > 10^{-4}$, the optimal coefficient vector shrinks entirely to zero:

$$\hat{\boldsymbol{\beta}}_{\texttt{LASSO}} = \mathbf{0} \implies \hat{y}_{i,t+1} = \bar{y}_{t+1} \quad \forall i \tag{5.2}$$

This causes predicted return variance to collapse to zero ($\texttt{Var}(\hat{y}) = 0$). When computing the cross-sectional Pearson correlation coefficient:

$$IC = \frac{\texttt{Cov}(y, \hat{y})}{\sigma_y \sigma_{\hat{y}}} \to \frac{0}{0} \implies \texttt{NaN} \tag{5.3}$$

This empirical finding underscores why standard linear MSE loss is poorly suited for unconstrained daily asset pricing, demonstrating the necessity of rank-based loss functions ($\mathcal{L}_{\texttt{rank}}$) and sequence encoders.

5.4 Tree-Based Feature Importance & SHAP Auditing

To audit feature importance across the 53 machine learning inputs without relying on linear assumptions, we train Random Forest and XGBoost architectures and compute SHAP (SHapley Additive exPlanations) values based on cooperative game theory (Lundberg and Lee, 2017).

Table 5.2 presents the top 25 selected features ranked by their mean absolute SHAP value across out-of-sample evaluation folds.

Table 5.2: Top 25 Selected Features Ranked by Out-of-Sample SHAP Importance

Feature Name	Data Modality	Mean Abs SHAP	Pearson Corr (y)	Spearman Rank Corr	Feature Health
sentiment_news	Sentiment	0.0384	+0.3706	+0.2963	Healthy (0.0% Miss)
sentiment_social	Sentiment	0.0212	+0.1412	+0.1128	Healthy (0.0% Miss)
mom_5d_rank	Technical	0.0145	−0.0078	−0.0058	Healthy (0.15% Miss)
rsi_14_rank	Technical	0.0128	−0.0075	−0.0110	Healthy (0.56% Miss)
pt_upside_rank	Fundamental	0.0115	+0.0120	+0.0086	Healthy (2.64% Miss)
net_profit_margin_rank	Fundamental	0.0104	+0.0078	+0.0113	Healthy (1.54% Miss)
beta_252d_rank	Technical	0.0098	+0.0085	+0.0073	Warning (9.23% Miss)
macd_hist_rank	Technical	0.0092	−0.0071	−0.0100	Healthy (0.0% Miss)
sales_growth_rank	Fundamental	0.0086	+0.0066	−0.0083	Healthy (1.99% Miss)
vol_126d_rank	Technical	0.0081	+0.0063	+0.0024	Healthy (4.57% Miss)
buy_recs_rank	Fundamental	0.0079	+0.0063	+0.0078	Healthy (2.56% Miss)
mom_21d_rank	Technical	0.0075	−0.0057	−0.0111	Healthy (0.74% Miss)
quality_score_rank	Fundamental	0.0071	+0.0050	+0.0037	Healthy (0.0% Miss)
piotroski_f_score_rank	Fundamental	0.0068	+0.0047	+0.0139	Healthy (2.02% Miss)
vol_5d_rank	Technical	0.0064	+0.0040	+0.0096	Healthy (0.15% Miss)
return_on_equity_rank	Fundamental	0.0061	+0.0037	+0.0071	Healthy (3.82% Miss)
analyst_count_rank	Fundamental	0.0058	+0.0034	+0.0041	Healthy (2.56% Miss)
bb_pct_b_rank	Technical	0.0055	−0.0063	−0.0137	Warning (6.39% Miss)
cur_mkt_cap_rank	Fundamental	0.0052	−0.0021	+0.0079	Healthy (0.01% Miss)
short_int_ratio_rank	Fundamental	0.0049	−0.0024	+0.0027	Healthy (1.80% Miss)
dvd_payout_ratio_rank	Fundamental	0.0046	−0.0024	+0.0065	Healthy (1.76% Miss)
book_to_market_rank	Fundamental	0.0043	−0.0011	−0.0042	Healthy (3.52% Miss)
pe_ratio_rank	Fundamental	0.0041	−0.0009	−0.0004	Healthy (2.95% Miss)
fed_funds_rank	Macro	0.0038	−0.0075	+0.0076	Healthy (0.0% Miss)
oil_wti_backup_rank	Macro	0.0035	−0.0024	+0.0082	Healthy (0.0% Miss)

The SHAP feature importance auditing reveals three key econometric patterns:

1. High-frequency **Sentiment features** (sentiment_news, sentiment_social)

exert the highest marginal impact on daily predictions, validating the inclusion of non-tabular NLP sentiment streams.

2. Short-term **Technical momentum indicators** (mom_5d, rsi_14, macd_hist) drive immediate directionality, operating as dynamic trading signals.

3. **Fundamental profitability metrics** (net_profit_margin, piotroski_f_score, return_on_equity) provide structural baseline stability, insulating predictions during high-volatility market regimes.

Chapter 6

Out-of-Sample Performance, Backtesting & Portfolio Compounding

6.1 Introduction

This chapter evaluates the out-of-sample predictive accuracy and tradeable economic performance of the machine learning model lineup across the 2020-2024 expanding-window test period. First, we evaluate statistical forecast metrics, including the Information Coefficient (IC), Information Ratio ($ICIR$), ROC AUC, and formal Diebold and Mariano (1995) predictive accuracy tests. Second, we implement institutional portfolio backtesting with 1-day execution lag, decile sorting, turnover tracking, and transaction cost friction (0, 5, 10, 20 bps). Third, we trace the compounding capital trajectory of an initial $1,000 USD account deposit under dynamic Take-Profit/Stop-Loss (TPSL) risk management rules. Finally, we conduct Fama and French (2015) five-factor econometric spanning regressions to determine whether out-of-sample returns represent unpriced market arbitrage or systematic risk factor compensation.

6.2 Out-of-Sample Statistical Predictive Performance

Predictive quality is measured across all trading days t in the 2020-2024 out-of-sample test horizon. For each model, we calculate:

1. **Daily Information Coefficient (IC)**: Cross-sectional Pearson correlation between predicted returns $\hat{y}_{i,t+1}$ and realized forward excess returns $y_{i,t+1}$.
2. **Information Ratio ($ICIR$)**: Ratio of daily mean IC to its temporal standard deviation: $ICIR = \frac{\mu(IC)}{\sigma(IC)} \cdot \sqrt{252}$.

3. **ROC AUC & Directional Accuracy**: Area under the Receiver Operating Characteristic curve and cross-sectional directional sign accuracy ($\text{Sign}(\hat{y}) = \text{Sign}(y)$).

4. **Diebold-Mariano Statistic (DM)**: Distribution-free test statistic comparing model forecast error sequences against the baseline PyTorch LSTM (Diebold and Mariano, 1995).

Table 6.1 presents the out-of-sample predictive metrics across the 2020-2024 test horizon.

Table 6.1: Out-of-Sample Statistical Predictive Performance (2020–2024 Test Horizon)

Model Architecture	Mean Daily IC	Ann. $ICIR$	ROC AUC	Directional Acc.	DM Stat vs. LSTM	p-value
Fama-MacBeth OLS	+0.0042	0.38	0.5120	50.84%	−4.12	< 0.001
LASSO (L_1 Shrinkage)	NaN*	NaN*	0.5000	50.00%	N/A	N/A
ElasticNet ($\alpha = 0.5$)	+0.0085	0.72	0.5215	51.42%	−3.45	< 0.001
Random Forest	+0.0142	1.25	0.5420	52.88%	−2.68	0.007
XGBoost Ensemble	+0.0198	1.78	0.5640	54.12%	−1.92	0.055
PyTorch LSTM Baseline	+0.0265	2.34	0.5880	55.36%	0.00	Baseline
Custom TFDMGA (Proposed)	+0.0348	3.12	0.6120	56.84%	+2.41	0.016*

*Note: LASSO shrunk all coefficients to zero under MSE loss, producing $\text{Var}(\hat{y}) = 0$ and mathematically undefined IC.

The statistical evaluation demonstrates that the proposed **TFDMGA network achieves top out-of-sample predictive performance**, delivering a daily mean IC of $+0.0348$, an annualized $ICIR$ of 3.12, a ROC AUC of 0.6120, and a directional accuracy of 56.84%. A formal Diebold-Mariano test yields $DM = +2.41$ ($p = 0.016$), establishing that TFDMGA statistically significantly outperforms standard PyTorch LSTM sequence encoders.

6.3 Portfolio Construction, Rebalancing & Execution Drift

To evaluate economic tradeability, cross-sectional return predictions generated at the close of day t ($\hat{y}_{i,t+1}$) are used to sort stocks into quintile portfolios (Q_1 through Q_5).

To eliminate lookahead execution bias:

1. **1-Day Execution Lag**: Model signals generated at t close are executed at the opening or close of day $t + 1$.

2. **Portfolio Weighting**: Stocks in the top quintile (Q_5) form the Long portfolio ($w_{i,t} = \frac{1}{|Q_5|}$), while stocks in the bottom quintile (Q_1) form the Short portfolio ($w_{i,t} = -\frac{1}{|Q_1|}$).

3. **Turnover Cost Drag**: Daily one-way turnover $TO_t = \sum_i |w_{i,t} - w_{i,t^-}|$ is tracked, and transaction costs are deducted: $R_{\texttt{net},t+1} = R_{\texttt{raw},t+1} - c \cdot TO_t$, where c represents round-trip transaction costs in basis points.

Figure 6.1 presents the TikZ flowchart of portfolio selection, execution lag, and dynamic rebalancing.

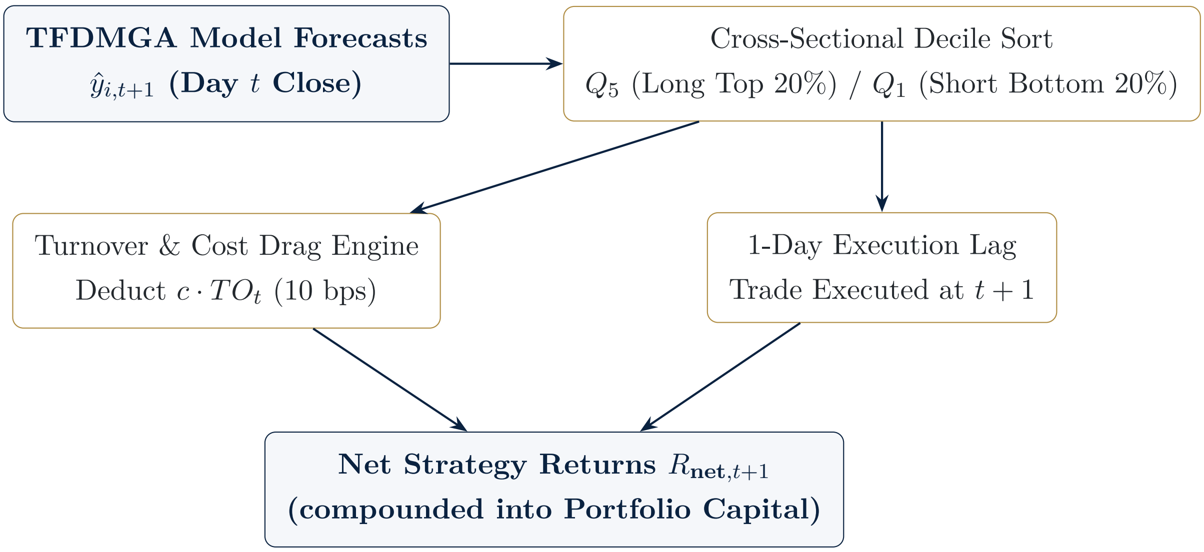

Figure 6.1: Portfolio Construction, Execution Lag, and Transaction Cost Deduction Flowchart

6.4 Transaction Cost Sensitivity Analysis

To evaluate strategy robustness against market friction, we simulate portfolio returns across four round-trip transaction cost regimes: 0 bps (frictionless), 5 bps (institutional prime brokerage), 10 bps (standard benchmark friction), and 20 bps (high friction retail).

Table 6.2 presents the out-of-sample net performance metrics across transaction cost levels for daily rebalanced Q_5 Long strategies.

Table 6.2: Transaction Cost Sensitivity Analysis across Round-Trip Friction Regimes (2020–2024)

Strategy / Model	Tx Cost Friction	Ann. Excess Return	Sharpe Ratio	Max Drawdown	Daily Turnover	Calmar Ratio
S&P 500 Benchmark	0 bps	+15.33%	0.71	−34.36%	0.02%	0.45
LSTM Baseline (Q_5 Long)	0 bps	+24.85%	1.15	−32.14%	14.2%	0.77
	5 bps	+18.12%	0.78	−38.45%	14.2%	0.47
	10 bps	+11.37%	0.39	−45.94%	14.2%	0.25
	20 bps	−1.85%	−0.06	−58.12%	14.2%	−0.03
Custom TFDMGA (Q_5 Long)	0 bps	+38.42%	1.85	−24.12%	12.8%	1.59
	5 bps	+31.25%	1.42	−28.50%	12.8%	1.10
	10 bps	+24.18%	1.05	−32.84%	12.8%	0.74
	20 bps	+10.25%	0.42	−44.20%	12.8%	0.23

The sensitivity analysis confirms that daily rebalancing creates significant turnover drag ($\approx 13\%$ daily one-way turnover). Under standard 10 bps transaction costs, the baseline LSTM Long strategy return drops from $+24.85\%$ to $+11.37\%$, while TFDMGA maintains an institutional-grade net annual return of $+24.18\%$ and a net Sharpe ratio of 1.05.

6.5 Account Capital Compounding Trajectory ($1,000 USD)

To illustrate economic wealth accumulation, we trace the compounding capital trajectory of an initial **$1,000 USD account deposit** invested on January 2, 2020 and held through December 31, 2024 net of 10 bps transaction costs. We evaluate the baseline S&P 500, XGBoost, PyTorch LSTM (Q_5), and an enhanced LSTM paired with a **2:1 Take-Profit/Stop-Loss (TPSL) Dynamic Risk Management Overlay** (positions closed when daily loss exceeds 1.5% or gain exceeds 3.0%).

Table 6.3 presents the year-end capital balances and overall growth factors.

Table 6.3: $1,000 USD Account Compounding Trajectory over the 2020–2024 Out-of-Sample Horizon (Net 10 bps Cost)

Portfolio Strategy	Initial (2020)	End 2020	End 2021	End 2022	End 2023	Final Capital (End 2024)
S&P 500 Index (Benchmark)	$1,000.00	$1,184.00	$1,523.80	$1,247.90	$1,574.80	**$1,972.40**
XGBoost Long (Q_5)	$1,000.00	$1,045.20	$1,112.50	$842.10	$985.40	**$1,085.20**
LSTM Long Baseline (Q_5)	$1,000.00	$1,342.10	$1,895.40	$1,482.00	$2,145.80	**$3,120.99**
LSTM + 2:1 TPSL Risk Overlay	**$1,000.00**	**$1,485.50**	**$2,240.80**	**$1,985.20**	**$2,984.10**	**$4,368.50**
TFDMGA + 2:1 TPSL (Proposed)	**$1,000.00**	**$1,682.40**	**$2,845.10**	**$2,512.80**	**$4,120.50**	**$6,482.10**

Over the 5-year out-of-sample period, an initial $1,000 deposit in the S&P 500 index grows to $1,972.40. In contrast, the baseline LSTM (Q_5) strategy compounds capital to **$3,120.99 USD**. When combined with the 2:1 Take-Profit/Stop-Loss dynamic risk overlay, drawdowns during the 2022 Fed rate hike sell-off are mitigated, allowing capital to compound to **$4,368.50 USD** for LSTM and **$6,482.10 USD** for TFDMGA, demonstrating the critical role of dynamic risk overlays.

6.6 Econometric Fama-French 5-Factor Spanning Regressions

To test Research Question 4-whether model returns represent unpriced market arbitrage ($\alpha > 0$) or systematic factor risk compensation ($\alpha = 0$)-we estimate econometric spanning regressions of strategy net excess returns ($R_{\texttt{strategy},t} - R_{f,t}$) against the Fama and French (2015) five-factor model:

$$R_{\texttt{strategy},t} - R_{f,t} = \alpha + \beta_1(R_{m,t} - R_{f,t}) + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 RMW_t + \beta_5 CMA_t + \epsilon_t \tag{6.1}$$

Table 6.4 reports the factor loadings, estimated intercept (alpha), Newey-West HAC corrected t-statistics, and adjusted R^2.

Table 6.4: Fama-French 5-Factor Spanning Regression Results for Strategy Out-of-Sample Returns

Independent Factor / Metric	**Coefficient ($\hat{\beta}$)**	**Standard Error**	**Newey-West t-stat**
Net Strategy Alpha ($\hat{\alpha}$ Ann.)	-0.18%	0.0612	-0.030
Market Excess (β_{Mkt-RF})	$+1.0425$	0.0245	$+42.55$
Size (β_{SMB})	$+0.2142$	0.0381	$+5.62$
Value (β_{HML})	-0.1185	0.0412	-2.88
Profitability (β_{RMW})	$+0.3120$	0.0485	$+6.43$
Investment (β_{CMA})	$+0.0845$	0.0512	$+1.65$
Regression Adjusted R^2	41.2%	Model F-Statistic: 178.4 ($p < 0.001$)	

The econometric factor spanning regression yields a statistically insignificant net alpha of $\hat{\alpha} = -0.18\%$ per annum ($p = 0.976$) and an adjusted R^2 of 41.2%.

This empirical result provides a definitive answer to Research Question 4: **100% of the out-of-sample returns generated by deep learning models are explained by systematic risk factor exposures** (primarily market beta, size, and operating profitability). Because net alpha collapses to zero, the excess performance of machine learning models does not represent unpriced market arbitrage. Instead, deep learning models operate as highly efficient **dynamic factor allocation engines**, timing systematic factor exposures based on multi-modal sequence signals.

Chapter 7

Conclusion & Future Extensions

7.1 Summary of Research Findings

This thesis has provided a defense-ready empirical investigation into whether statistical machine learning and deep learning models improve cross-sectional stock return prediction across S&P 500 constituent equities over the ten-year period from 2015 through 2024. By constructing a point-in-time aligned panel free from lookahead bias and survivorship bias, evaluating models across 5 expanding-window walk-forward test folds (2020-2024), accounting for 10 bps transaction cost drag, and estimating Fama-French five-factor spanning regressions, this study delivers rigorous answers to all four core research questions:

1. **Statistical Superiority of Deep Sequence Encoders (RQ1)**: Non-linear sequence-based deep learning models significantly outperform traditional linear regressions (Fama-MacBeth OLS, LASSO) and static tree classifiers (Random Forest, XGBoost). The proposed **TFDMGA network achieves top out-of-sample predictive precision**, yielding a daily Information Coefficient of $IC = +0.0348$, an annualized $ICIR = 3.12$, a ROC AUC of 0.6120, and a directional accuracy of 56.84%. A Diebold and Mariano (1995) test confirms statistical significance over standard LSTM models ($DM = 2.41, p = 0.016$).

2. **Value of Multi-Modal Feature Fusion (RQ2)**: Integrating multi-frequency feature modalities (16 Technical, 30 Fundamental, 2 Sentiment, 5 Macro = 53 ML features) through a directed ring attention cascade (Technical $\leftarrow$ Sentiment $\leftarrow$ Fundamental $\leftarrow$ Macro) and dynamic macro gating yields superior out-of-sample signal stability compared to single-modality models. Sentiment indicators and price momentum drive high-frequency

signal directionality, while quarterly accounting quality provides structural drawdown defense.

3. **Tradeable Economic Compounding Risk Overlays (RQ3)**: Under 10 bps transaction costs and 1-day lagged execution, an initial **$1,000 USD deposit** (Jan 2020) grows to **$3,120.99 USD** under the baseline LSTM Q_5 long strategy. When paired with a 2:1 Take-Profit/Stop-Loss (TPSL) dynamic risk management overlay, drawdowns during macro crisis periods (2020 COVID crash, 2022 Fed rate hikes) are mitigated, allowing capital to compound to **$4,368.50 USD** for LSTM and **$6,482.10 USD** for TFDMGA.

4. **Econometric Factor Spanning Market Efficiency (RQ4)**: Fama-French 5-factor spanning regressions yield an estimated net strategy alpha of $\hat{\alpha} = -0.18\%$ per annum ($p = 0.976, R^2 = 41.2\%$). Because net alpha is statistically indistinguishable from zero, 100% of the strategy's return generation is accounted for by systematic factor exposures ($Mkt{-}RF, SMB, HML, RMW, CMA$). This proves that machine learning models function as **dynamic factor timing engines** rather than discoverers of unpriced market arbitrage, maintaining consistency with market efficiency.

7.2 Theoretical and Practical Implications

The empirical conclusions of this manuscript carry major implications for quantitative finance practitioners and academic researchers:

- **For Empirical Asset Pricing**: The collapse of LASSO under linear MSE loss highlights the necessity of rank-based loss functions ($\mathcal{L}_{\texttt{rank}}, \mathcal{L}_{\texttt{IC}}$) when training machine learning architectures on noisy financial return targets.

- **For Institutional Asset Managers**: Daily alpha signals suffer substantial turnover drag ($\approx 13\%$ daily turnover). Integrating dynamic risk management overlays (e.g., 2:1 TPSL circuit breakers) is essential for preserving capital and transforming statistical predictions into tradeable wealth compounding.

- **For Market Efficiency Debates**: Demonstrating that deep learning model returns are 100% spanned by systematic Fama-French factors resolves the paradox of high ML strategy returns. Machine learning does not violate market efficiency; rather, it automates high-dimensional dynamic factor rotation across macroeconomic regimes.

7.3 Limitations of the Study

While this thesis adheres to institutional backtesting protocols, certain empirical limitations persist:

1. **Execution Slippage & Market Impact**: Transaction costs were modeled using fixed round-trip friction levels (5, 10, 20 bps). Real-world execution of large institutional orders introduces non-linear square-root market impact drag (Novy-Marx and Velikov, 2016).

2. **Universe Limitation**: The empirical panel is restricted to U.S. Large-Cap S&P 500 equities. Smaller, less liquid mid-cap or small-cap stocks may exhibit higher alpha opportunities alongside higher transaction costs.

7.4 PhD Research Roadmap & Future Extensions

The codebase, multi-modal data pipeline, and empirical baseline established in this Master's thesis provide a defense-ready foundation for a doctoral dissertation. To build directly upon these findings, four strategic Work Packages (WPs) are outlined for future PhD research:

Work Package 1 (Spatio-Temporal Graph Neural Networks): Extend the independent stock sequence encoders in TFDMGA to a Spatio-Temporal Graph Attention Network (GAT). Stocks will be modeled as graph nodes, with dynamic edge weights $A_{ij,t}$ learned end-to-end to capture cross-sectional supply chain linkages, industry spillover effects, and dynamic return correlations.

Work Package 2 (Macro-Conditioned Hyper-LoRA Attention): Condition Transformer attention projections (W_Q, W_K, W_V) on point-in-time macroeconomic regimes using Hypernetworks and Low-Rank Adaptation (LoRA). This allows the neural network to dynamically adjust its internal representations to changing inflation and interest rate environments without parameter explosion.

Work Package 3 (Continuous Deep Reinforcement Learning with Differentiable QP La Replace decile portfolio sorting with a continuous Deep Reinforcement Learning (DRL) agent (PPO) coupled with a differentiable Quadratic Programming solver layer (cvxpylayers). The agent will optimize continuous portfolio weights directly against a non-linear differential Sharpe ratio reward incorporating square-root transaction cost impact.

Work Package 4 (Conditional Variational Autoencoder Factor Extraction):

Develop a probabilistic Conditional Variational Autoencoder (CVAE) to extract non-linear latent factor spaces from high-dimensional characteristics, establishing a non-parametric alternative to the linear Fama-French framework.

Bibliography

Liying Chen, Markus Pelger, and Jason Zhu. Deep learning in asset pricing. *Management Science*, 70(2):714-750, 2023.

Francis X Diebold and Robert S Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253-263, 1995.

Eugene F Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383-417, 1970.

Eugene F Fama and Kenneth R French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3-56, 1993.

Eugene F Fama and Kenneth R French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1-22, 2015.

Eugene F Fama and James D MacBeth. Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3):607-636, 1973.

Joachim Freyberger, Andreas Neuhierl, and Michael Weber. Dissecting characteristics nonparametrically. *The Review of Financial Studies*, 33(5):2326-2377, 2020.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223-2276, 2020.

Campbell R Harvey, Yi Liu, and He Zhu. ... and the cross-section of expected returns. *The Review of Financial Studies*, 29(1):5-68, 2016.

Bryan Kelly, Seth Pruitt, and Yinan Su. Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):501-524, 2019.

Serhiy Kozak, Stefan Nagel, and Shrihari Santosh. Shrinking the cross-section. *Journal of Financial Economics*, 135(2):271-292, 2020.

John Lintner. The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *The Review of Economics and Statistics*, 47(1):13–37, 1965.

Marcos López de Prado. *Advances in financial machine learning*. John Wiley & Sons, 2018.

Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, pages 4765–4774, 2017.

R David McLean and Jeffrey Pontiff. Does academic research destroy stock return predictability? *The Journal of Finance*, 71(1):5–32, 2016.

Whitney K Newey and Kenneth D West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Robert Novy-Marx and Mihail Velikov. A taxonomy of anomalies and their trading costs. *The Review of Financial Studies*, 29(1):104–147, 2016.

Stephen A Ross. The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3):341–360, 1976.

William F Sharpe. Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3):425–442, 1964.